AI and Cybersecurity Research Portfolio

Prepared for portfolio purposes by Isaac. Proposed designs and illustrative scenarios are clearly distinguished from verified work experience.

Abstract

Research-style report examining AI in cybersecurity.

Integrity Statement

Survey figures are illustrative and require source verification.

Background

Growth of AI-assisted security operations.

Literature Themes

Threat detection, automation, anomaly analysis.

Illustrative Survey Data

Example portfolio figures.

Metric	Result
Detection improvement	72%
Training need	83%
Privacy concern	61%

AI for SOC

Automation of triage and alert correlation.

Phishing Detection

NLP-assisted message analysis.

Malware Classification

Model-driven categorization.

Risks

Bias, poisoning, hallucination and privacy issues.

Risk	Description
Poisoning	Corrupt training data
Hallucination	Incorrect output

Governance

Human oversight and audit trails.

Recommendations

Validation, testing and compliance review.

Conclusion

AI augments rather than replaces analysts.